

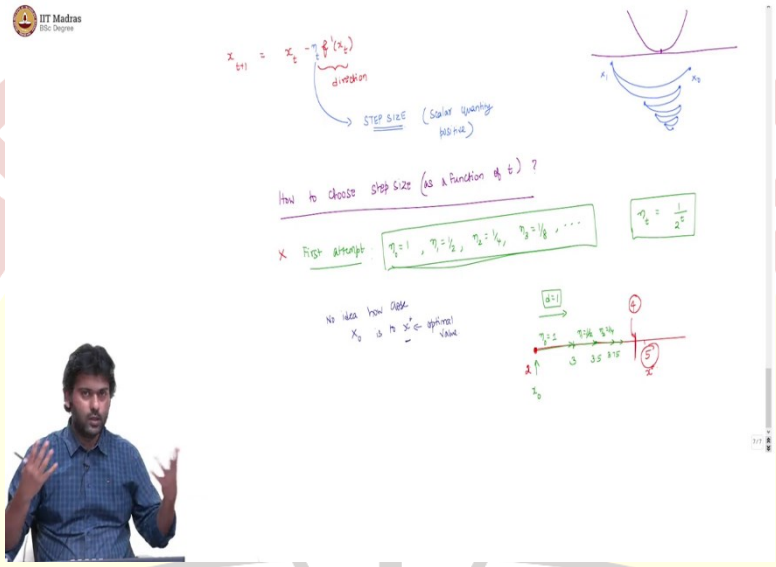

# IIT Madras

## ONLINE DEGREE

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**Solving an Unconstrained Optimization Problem (Part 2)**

I am sure most of you must have guessed, what is the issue, the issue is the amount and how do you fix the amount.

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Well, one way to do this would be to do the following. So, your update rule x t + 1, right now says it is x t - f ‘ of x t. So, this is the direction, which we think is the right way to move. Of course, we have not argued this for general functions, we are just saying this is true for x - 5 squared, but we will come to that generalization a little bit later, for the moment assume that there is a good direction, which means that the direction is not the issue, it is more the issue of how much you are moving in this direction.

One way to fix this would be to decide how much you want to move in this particular direction. So, and that we will call as a new value, which we will multiply. So, let me make this clear. So, that we do not, there is no confusion, we are going to multiply a scalar value, which we are going to call as the step size.

So, you want to take one step in the direction of - f ‘ of x t, but not all the time. So, you want to take smaller and smaller steps potentially and that is what this $\eta$ t is going to do. So, this is a value, it is a scalar, it is a scalar quantity and dispositive. Well, if $\eta$ t equals 1, for every t, then it means that whatever we had earlier, we would go back to that. But that is a bad

algorithm, we know that there is an example where if you start with 10, you are going to oscillate. So, $\eta$ t cannot be 1 all the time.

Now, which means that the idea is that you start maybe with 1, so you start from some x 0, maybe you start with 1 because you do not know what is a good step size, maybe you start with 1, you will go somewhere else. Now, maybe you should not take a step of one again from this new point, maybe you will take only a smaller step.

So, maybe you will take a smaller step, maybe you will come here and then you will take even smaller step and if you keep doing that, hopefully you will converge to the minimum. So, of course, I am talking about this particular function that we have. So, that is the idea. So, you do not want to take the same steps size in each round. But you want to somehow minimize this.

Now, the question is how to choose the step size. It seems like a reasonable idea to choose different step sizes, in different rounds and it also seems like a reasonable idea to make this smaller and smaller, so that we do not really, end up oscillating, so that at least tries to solve that problem hopefully, but it is not clear eight how to choose the step size.

So, as a function of t. I mean, the step size is going to be different for different iterations and we want like a decreasing sequence of step size. So, ideally as a function of t. The question then is that to complete our algorithm, we also have to specify how to choose the step size. Now, there are several ways to reduce your step size.

So, you can start with one and then make it any number smaller than this in different ways. So, there are so many sequences of step sizes that one might want to, one might be able to choose, of course. Now, the question is, what is a good sequence of step sizes? How do we determine that? Well, let us take one example and see if that makes sense. So, here is an first attempt at coming up with a step size sequence.

Well, here is a sequence. So, $\eta$ well $\eta$ 0 is 1 $\eta$ 1 is half, $\eta$ 2 is a quarter $\eta$ 3 is one eight, and so on. In general, you will have $\eta$ t as 1 by 2 to the t. Basically, you are halving your step size that every time. It starts with 1 then becomes half then becomes quarter and so on and it decreases as t increases, which seems, which seems like a reasonable first attempt.

Now, what might be, so the question to ask, is this the best possible step size? I mean, is it a good step size? If not, are there any potential issues that we might have sidestepped, it by using the step size. Well, one thing to keep in mind is the following. So, your x 0 that you are starting with, you have no clue how close you are to the actual minimum, so nobody tells you where

the actual minimum is at least when you are running it as a computer, the computer has no clue where the actual minimum indeed is.

It starts with an arbitrary potential x 0 and when it starts with an arbitrary x 0, it is it is not clear how close is x 0 to x star, where x *let us say is the minimum. So, no idea how close x 0 is to x star, so x is the optimal value. Now, which means that you might be really, really far off from your x *and if you keep doing this iterative algorithm by taking the negative of the derivative and taking $\eta$ t size in that particular direction. If you keep repeating this, you should eventually hope that you will reach x *that is the hope.

So, we want to get to x star. But we do not know how close x 0 is to x star, which means that if you are taking steps in directions, the step sizes should not really become too small after a point that maybe you are still in the right direction. But your step sizes are too small that you are not even able to reach x star. That is a problem that might happen and we should be very of that.

So, for example, in this particular step size that we have chosen, let us say if I start from a particular value and for the moment assume that my function is such that it may not be x - 5 square, it is a different function, which will always give me d equals 1. So the direction is always 1 unit in the positive direction. Let us say the function is like that.

I mean, if you want to think about what function would have the direction always equal to 1, feel free to think about that. But that is not the main point here. I am just saying that let us say there is a function, no matter which point you ask for the direction, you take the - f ‘ of x, it is always going to give you a value of 1.

So, let us say we start and let us say the optimum value x *is 5. So, for this particular function. Now, let us say we start at 2, let us say we start at the point 2 and we need to reach 5, which is where the optimal is. So, this is where the optimal is, this is x star. Now, what is my algorithm going to do, what I am saying is that this is x 0 for our algorithm and I have to reach x *and if I use the step size, which we just came up with, which is $\eta$ t is 1, 1 over 2 to the t, which means that in the first round, I am going to take a step of one unit.

So, from 2 I will take a step of 1 unit and I will reach 3, which is good, I am getting closer to the minimum. Now in the second round, I am going to take you half step, which means from 3 this is a 1 unit, $\eta$ 0 was 1 and d is always 1, so which means that 1 into 1 I take a took a step of

1 here $\eta$ 1 is going to be half, because it is 1 over 2 power t which means that I will end up with 3.5.

Now $\eta$ 2 is going to be quarter, $\eta$ 2 is going to be quarter which means I will reach 3.75. Now, if I keep doing this think where you will eventually end up. So, will you reach 5 or not? Well, as you can see, you are getting smaller and smaller step sizes. You started with 2, I mean there was a good step size initially from 2 to 3, then it is becoming smaller and smaller every round.

Now, if you do, I mean if you think about this for a minute, you will understand that no matter how many rounds you run this algorithm, even if you run this for infinite amount of time, you are never going to get any closer to 5 than 4. At every round, your derivative, negative derivative was giving you the right direction, which was always 1.

So, it was always saying that move to your right. Which was the correct direction. But our step sizes we designed it such that it kind of reduced by half at every round and unfortunately, that is only leading us till the value of 4, we will never reach the value of 5. Now, this is the opposite problem of what we had earlier.

The problem that we had earlier was you took, if you did not have step sizes, you might take too large step in the direction of the gradient, in the direction of the negative derivative and that will be problem because it might lead to oscillations. Now, we introduce a step size and we see that, well if you are not careful about choosing the sequence of step sizes, it might result in the opposite problem, where your directions are correct, but then you are not eventually reaching the minimum, because your steps are too small in that direction, which is also a problem to be very of.

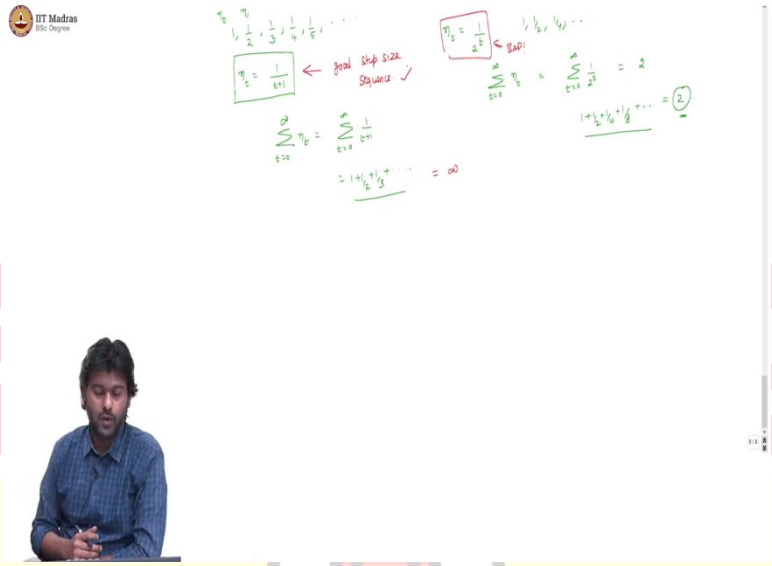
Because we will not then be able to reach 5. Now, the question is which means that this first attempt is a failure. So, the question is, if this sequence of step sizes is not a great sequence to actually use, then what else should we do? So, how well can we come up with step sizes? Is there a different sequence that we can think of which might not have this problem.

So, the problem is that you might, you have two conflicting objectives. One is, you want to reduce your step size as you increase your number of iterations, otherwise it might lead to oscillations. The other objective is that you should not reduce it too much that you will not be able to reach your minimum.

So, which means that your reduction should still be such that you will be able to reach your minimum no matter how far your minimum is from the place where you started. Now, is there

such a step size at all, I mean these look like conflicting objectives is there such as step size at all, is something that you can pause and think and we will talk about this next.

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So, now I am going to come up with a new sequence of step sizes, different from the previous step size that we looked at and it is going to look as follows. So, I am going to think of my n t as follows. Maybe I will give you the sequence and then we will generalize it. So, my sequence is going to be 1 by 2, so far same as what we had earlier.

Now, the third step in the previous case was 1 over 4. Now, I want to say it is 1 over 3 and this is 1 over 4, 1 over 5. Which means that my $\eta$ t, if I want to write this formally, because $\eta$ t starts from, this 0, this is $\eta$ 1, this is $\eta$ 2 and so on. So, my $\eta$ t is going to be 1 by t + 1. So, this is my new sequence of step size.

At first glance, it seems that this should have the same issue as the previous sequence that we tried, which was 1 over 2 power t, because the values 1 by 2, 1 by 3 and so on, also seem to kind of go towards 0. So, which means that, well it might seem that I might not really be able to, reach there might be cases where I will not be able to reach the minimum.

So, in other words, if you think about this, in the previous case, $\eta$ t was 1 over 2 power t. So, which was 1 half one fourth,… and this sequence had the property that if some t equals 0 to infinity $\eta$ t which is the sum of all my step sizes. So, if we start from a particular point and they summed up all my step sizes, let us say my D was always 1. So, then this is t equals 0 to infinity 1 over 2 power t of course, 1 over 2 power t.

If I did this sum, well the sum is going to converge to the value 2, why this is 1 + half, + quarter + one eight and this is one can show that this sequence will actually be good too. So, which means that no matter where I start, if I choose this sequence and if my direction is the same, then I would not be able to get too far. So, I cannot get to more than 2 units and this is a bad thing.

So, because we cannot move too much. Now, what happens to this case? So, if I do t equals 0 to infinity, $\eta$ t here, so which becomes t equals 0 to infinity, 1 over t + 1, which is 1 + 1 by 2 + 1 by 3, + ., ., .. Now, the question is, where does this converge to? So, this sequence went to 2. What about this sequence?

Well, it turns out that this sequence does not converge to anything. So, this actually is infinity. So, it is a very, very interesting sequence, the values become smaller and smaller at the cumulative sum grows to infinity. So, which means that no matter where I start, because my step sizes, though they get smaller, because the cumulative step sizes is going to go to infinity, I will be able to make sure that I will eventually reach the minimum.

Of course, there are a lot of technical details to show that this is indeed the case and so on. But the point I want to make is that this is a good step size sequence. This rather, this bad, does not work, whereas this typically works. So, without going too much details about why this works and so on. This is the intuition as to why it works, because this cumulative sum is actually infinity.